

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EX-405 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Linear Integrated Circuits and its Applications
Time : Three Hours

Maximum Marks : 70

Note: i) Answer any five questions.

ii) All questions carry equal marks.

1. A square wave with negligible rise time at a peak to peak amplitude of 500 mV must be amplified to a peak-to-peak amplitude of 3V, with a rise time of μ sec or less. Can a 741 with a slew rate of $0.5V/\mu$ sec be used?
2. Write different characteristics of filters. Give classification of filters used in OP-AMP. Explain any one of them.
3. Define the following terms:
 - i) CMRR
 - ii) Input offset current
 - iii) Input offset voltage
 - iv) Slew Rate will differentiate an input signal with $f_{max} = 100$ Hz.
4. Explain the term input bias current. Explain how the effect of input bias current on the output offset voltage reduces when a compensating resistance is connected between the ground and non-inverting input.
5. Draw the block diagram of Operational Amplifier. Why we need d.c. level shifting current? Explain it.
6. What are the main characteristics of a Butterworth filter? What are the applications of a Butterworth filter.

OR

Design a first order low pass filter for the following specifications

- i) Pass band voltage gain = 2
 - ii) Cut-off frequency = 10kHz.
7. Design and Explain:
- a) Differentiators circuit
 - b) Log amplifier
- OR
- Write short notes on (any four)
- a) Analog multiplier - MPY634
 - b) TL082 datasheet
 - c) Log/Anti log amplifier
 - d) Current to voltage converter
 - e) Notch filter
 - f) Triangular/ Rectangular wave Generator
8. Write short notes (any two):
- a) PLL
 - b) Voltage Regulators
 - c) Dual Power Supply
 - d) Notch filter

EX-405 (GS)